INCH-POUND

MIL-PRF-32794
30 July 2025

PERFORMANCE SPECIFICATION

High Temperature Conductive Inks for Additive Manufacturing of Electronic Components

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for heat-resisting, cured/sintered conducting inks for printed electronics that will withstand a minimal temperature of 572 °F (300 °C) in an oxygen rich environment. These conductive inks are used for Inkjet Printing, Aerosol-jet Printing, Extrusion Direct Writing Printing, and other types of additive manufacturing processes. These inks are intended to print on dielectric surfaces that are exposed to high temperatures in an oxidative environment. The cured inks must be electrically conductive at high temperature and have high acceleration shock tolerances.

1.2 Product Classification. The material will be of the following types and classes as specified (see 6.2).

1.2.1 Types. The following types are defined by inks.

1.2.1.1 Type I. Type I is liquid ink.

1.2.1.1.1 Type IA. Type IA is particle suspension ink.

1.2.1.1.2 Type IB. Type IB is a particle-free solution ink.

Comments, suggestions, or questions on this document should be addressed to Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate (ARD), DEVCOM ARL Specifications & Standards Office, ATTN: FCDD-RLA-MC, Aberdeen Proving Ground, MD 21005-5069. Since contact information may change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9630

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1.2.1.2 Type II. Type II is a paste ink.

1.2.2 Classes. The following classes are defined based on service temperatures under the following material limitations or candidates.

1.2.2.1 Class I. Class I at 572 °F (300 °C) is for extremities of typical space travel, using flexible circuits in its hardware.

1.2.2.2 Class II. Class II at 932 °F (500 °C) is for atmospheric reentry tolerance.

1.2.2.3 Class III. Class III at up to 1,832° F (Up to 1,000 °C) is the super material in sensor devices.

1.2.2.4 Class IV. Class IV for above 1,832 °F (Above 1,000 °C) is the ultimate material and is used in solar expeditions.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are referenced in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ASTM INTERNATIONAL

ASTM B193	-	Standard Test Method for Resistivity of Electrical Conductor Materials

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ASTM D2485	-	Standard Test Methods for Evaluating Coatings for High Temperature Service
ASTM D3163	-	Stand Test Method for Determining Strength of Adhesively Bonded Rigid Plastic Lap-Shear Joints in Shear by Tension Loading
ASTM D3359	-	Standard Test Methods for Rating Adhesion by Tape Test
ASTM D3924	-	Standard Specification for Standard Environment
ASTM D4496	-	Standard Test Method for D-C Resistance or Conductance of Moderately Conductive Materials
ASTM D4541	-	Pull-Off Strength of Coatings using Portable Adhesion Testers

(Copies of these documents are available from https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

(ISO) /ASTM 52921	-	Standard Terminology for Additive Manufacturing - Coordinate Systems and Test Methodologies
ISO 4624	-	Paints and varnishes - Pull-off test for adhesion
ISO 9000	-	Quality Management Systems - Fundamentals and Vocabulary

(Copies of these documents are available from https://www.astm.org.)

(ISO Standard: Available from American National Standards Institute (ANSI), https://www.ansi.org.)

INSTITUTE FOR PRINTED CIRCUITS (IPC) INTERNATIONAL, INC. / THE ASSOCIATION CONNECTING ELECTRONICS INDUSTRIES STANDARD / JAPAN ELECTRONICS PACKAGING AND CIRCUITS ASSOCIATION (JPCA)

IPC TM-650	-	Test Methods Manual

(Copies of this document are available from IPC Store - https://shop.ipc.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. All inks used shall be as specified in the contract, purchase order, or by the procuring activity and supplied under all provisions of this specification for chemical composition and

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quality program requirements (see 6.2). Minimum acceptance and other requirements shall be met as specified in this specification. Materials shall be an excellent conductor of electricity. Base substrates shall be specified. Silver or copper or combination inks shall be as designated in the contract. Dielectric inks shall be supplied ready for application at elevated temperatures by different methods. The material supplied under contract shall be identical, within manufacturing tolerances, to the product receiving inspection approval.

3.2 Feedstock reuse. Recycled or recovered materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements prior to use in a production run, such as a first article test, and promotes economically advantageous life cycle costs. Safe disposal shall be in accordance with the state and local environmental regulations.

3.3 Chemical composition. Single or multiple ingredients or material choices shall contain multi-metal combinations and shall conform to the chemical requirements testing specified in 4.7.1. Silver migration consideration if needed shall be written in the contract or purchase order. The material shall be stable with the purity and toxicity specified in the Safety Data Sheet (SDS).

3.4 Application. The printing method shall be as specified by the procuring activity. High radio frequency and low voltage operation applications shall be described in the technical product data sheet for its intended use. High temperature applications shall be recommended in accordance with the supplier's instructions during curing. For application properties and inspection purposes, the trace shall form an adhering, non-drip, tough film, and be freed from flaws or other defects that will reduce serviceability or be detrimental to appearance.

3.5 Sintering/curing and heat treatment technique. The ink curing requirement, such chemical cure, UV cure, and thermal cure shall be specified by the vendor. The technical product data. sheet on curing procedure shall be supplied by the vendor. When inspected as specified in 4.8.1, the cured ink shall show no flaking or chipping or delamination after being exposed to the heat exposure cycle. Post processing and subsequent treatment shall be enabling the components to meet the requirements of this specification.

3.6 First article. When specified (see 6.2), a statement showing the product analysis in complete detail, all material properties and chemical composition information shall be furnished to the purchaser. A sample shall be subjected to the first article inspection and tests in accordance with 4.2 (see 6.2).

3.7 Quantitative requirements - characteristics. High electrical conductivities and conductive versus temperature and shock (single shock event and cumulative) requirements for different applications shall be specified for their intended use (see 6.2). Coefficient of Thermal Expansion (CTE) shall match the application. Maximum particle size, surface tension, color appearance, and sintering time considerations, if needed, shall be specified in the contract or purchase order. Storage stability shall be within one year from the manufactured date. Adhesion inspection for both sheer and pull off shall be performed in accordance with ASTM D3359 and D4541 respectively.

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3.7.1 Condition in container. The ink in a freshly opened full container shall show no curdling, livering, caking, lumps, or skins and shall show no more settling than may be redispersed with a mixer to a homogeneous state when tested as specified in 4.8.2.

3.7.2 Storage stability. For applications requiring elevated temperature properties, the contractor shall demonstrate by presenting Objective Quality Evidence (OQE) or conducting special tests that storage stability and handling of inks shall adhere to environmental compliance for handling and disposal of supply.

3.7.2.1 Accelerated storage stability. There shall be no curdling or hard dry caking, and any sedimentation shall easily mix back into a smooth homogeneous state.

3.8 Qualitative requirements. The contract or purchase order shall include a quality assurance section in the quality control / quality assurance plan as specified in 4.9.1 (see 6.2).

3.8.1 Adhesion. The assessment of the adhesion of the cured ink on the substrate shall be determined by its ability to not peel from the substrate as specified in 4.9.2 when tested in accordance with both ASTM D 3359 and D 4541. After bake cured, the resultant test rating shall be classified as scale 4B or better, when tested as specified in 4.9.2. The pull-off strength shall exceed 6 megapascals when tested in ASTM D 4541.

3.8.2 Printability. Printing requirement shall be vendor specific requirement. The technical product data sheet on printing instruction shall be supplied by the vendor. Compatibility with various substrates and ink formulations shall be determined. Appearance of cured ink, as specified by the manufacturer, shall show no signs of visible defects and shall have a smooth finish, free from particles. The viscosity shall allow for printability.

3.8.3 Curing properties. The ink, when applied to component board as specified in 4.10, shall form a sticking, hard, tough film free from streaks. The heat exposure cycle shall be specified by the supplier. Accelerated high temperature exposure shall be used in testing. The substrate material and the conductor shall be chemically compatible.

3.9 Ink classes requirements. Class 1 (300 °C), class II (500 °C), class III (1000 °C), and class IV (1500 °C) – the classes shall have the following performance in electrical conductivity, percentage change of electrical conductivity after 100 hours in 21% oxygen, adhesion strength to substrate, and single lap shear test results when evaluated as outlined for the following tests in 4.7.2, 4.7.3, and 4.7.4, and 4.7.5 respectively.

3.10 Electrical conductivity, classes I, II, III, and IV. The performance of inks shall be as follows: The electrical conductivity shall be not less than 1,000,000 Siemens per meter (S/m) at 300 °C for class I. The electrical conductivity shall be not less than 100,000 Siemens per meter (S/m) at 500 °C for class II. The electrical conductivity shall be not less than 75,000 Siemens per meter (S/m) at 1000 °C for class III. The electrical conductivity shall be not less than 50,000 Siemens per meter (S/m) at 1500 °C for class IV.

3.11 Percentage (%) change, classes I (300 °C), II (500 °C), III (1000 °C), and IV, (1500 °C). The percentage (%) change of electrical conductivity for all classes shall be as follows: After 100 hours in ambient air (21% oxygen) shall be at a specified maximum of 1%.

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3.12 Adhesion strength to substrate, classes I, II, III, and IV. The performance of inks shall be as follows: Adhesion strength to substrate shall be greater than 2 x 10^6 Pascals (Pa) at 300 °C for class I, and at 500 °C for class II, and at 1000 °C for class III, and at 1500 °C for class IV. Adhesion at room temperature, the trace shall be rated at not less than 4B in accordance with ASTM D 3359.

3.13 Single lap shear test, classes I (300 °C), II (500 °C), III (1000 °C), and IV, (1500 °C). The performance of inks shall be tested for all classes at room temperature, when evaluated in accordance with ASTM D 3163, single lap shear testing shall result in a minimum of 2 x 10^6 Pascals (Pa) (see 4.7.5).

3.14 Workmanship. Component produced under this specification shall be uniform in quality and clean, smooth, and sufficiently free from delamination and other defects which may affect its use, detrimentally affect the serviceability for electronic parts. All details of workmanship shall conform to the best practice for high quality 3 D printing. Any additional surface trace requirements shall be specified by procuring activity or purchase order or as directed by the contracting officer (see 6.2).

3.15 User instruction markings. Legibly marked or labeled, and any special markings shall be specified by the contract or purchase order (see 6.2).

3.16 Safety Data Sheet (SDS). A SDS for the ink used by this specification shall be included with each shipment and submitted to the end user, contracting activity, and other pertinent government agencies as stated in FED-STD-313.

3.17 Toxic ingredients. Toxicity clearance - The ink shall have no adverse effects on the health of personnel when used for its intended purpose.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements and any supplementary requirements specified herein are classified as follows:

a. First article inspection and tests (see 4.2).

b. Part quality inspection/Production acceptance (see 4.5).

4.2 First article inspection and tests. When specified (see 6.2), the feedstock supplier shall furnish a test sample or samples for the first article inspection and approval. The supplier shall comply with this requirement at the time of the purchase order or contract with the procuring activity and at any time as agreed upon by the part supplier and contracting agency. Test results shall be provided for the procuring activity and shall comply with the purchaser's requirements.

4.3 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2k), the supplier shall be responsible for the performance of all inspection requirements (examinations, quality conformance, and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own ISO 9000/IEC 17025 or NADCAP certified facility or any other facilities certified to ISO 9000/IEC17025 or NADCAP for the

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performance of the inspection requirements specified herein. Daily inspection events which contribute to the overall monthly Objective Quality Evidence (OQE) need not to be performed by an ISO 9000/IEC 17025 or NAPCAP certified facility. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.4 Inspection conditions. Unless otherwise specified in the purchase order or contract, all test specimens shall be conditioned and tested in standard conditions.

4.4.1 Standard inspection conditions. Unless otherwise stated in the test method or paragraph, all inspections shall be performed in accordance with ASTM D3924, and standard condition shall be 73 ± 4 °F (23 ± 2 °C) and a 40 – 70 percent relative humidity.

4.4.2 Other conditions. Tests shall be conducted at the temperature of the specific class as defined as classes I, II, III, and IV (see 1.2.2).

4.5 Part quality inspection/production acceptance. All feedstocks / inks shall meet all requirements of section 3 for procedures listed in section 4.9. The inspections set forth in this specification shall become a part of the suppliers' overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the supplier of the responsibility of ensuring that all components or parts submitted to the Government for conformance and acceptance comply with all requirements of the contract or purchase order.

4.6 Sampling for conformance tests. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements. The purchaser shall select the invocation of a sampling standard. Samples from each inspection lot from production shall be selected at random to verify compliance with this specification. However, this does not authorize submission of known defective components, either indicated or actual, nor does it commit the Government to accept defective components.

4.7 Testing responsibility and facilities. Unless otherwise specified in the contract or purchase order, the supplier shall be responsible for the performance of all the requirement tests as specified herein unless a requirement is specifically different in the contract (see 6.2).

4.7.1 Chemical composition, inorganic compounds. Multi-metal combinations shall be confirmed by chemical requirements testing conducted by the vendor. Material choices shall be stable and have their accompanying SDSs available (see 3.3).

4.7.2 Electrical conductivity. The conductor materials shall be tested using the four wire Kelvin measurements for resistance measurement in accordance with any of the following standards: ASTM D4496 or ASTM B193 or IPC TM-650 (see 3.10).

4.7.3 Percentage change. For the calculation of percentage change in electrical conductivity, the following formula shall be used: ((final value - initial value) / initial value) * 100 (see 3.11).

4.7.4 Adhesion strength to substrate. A tape test shall be used to access adhesion by making cuts in the trace and applying and removing pressure sensitive tape as specified in ASTM D3359. The

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methods for determining the adhesion of cured trace by performing a pull-off test as specified in ASTM D 4541 / ISO 4624 (see 3.12).

4.7.5 Single lap test. For all classes, the test shall be conducted at room temperature (see 3.13).

4.8 Application method. Methods for inspection and acceptance of procedures shall be determined by the purchaser. Cured ink specimens shall be resistant to flexing and creasing. Visual inspection methods shall be used, and other requirements and/or special conditions shall be specified in the contract or purchase order (see 6.2).

4.8.1 Sintering/curing and heat treatment technique. The method and procedures shall be specified by the vendor (see 3.5).

4.8.2 Condition in container. Upon opening a full previously unopened container, the condition of the ink shall be examined for compliance with 3.7.1. Reseal, then agitate the container on a shaker machine and reexamine for compliance with 3.7.1. On reexamination of the ink, the disclosure of any gel bodies or undispersed pigment indicates unsatisfactory settling properties.

4.9 Test procedures. The storage stability method shall be used as specified by the vendor and after testing, the ink shall meet all the requirements of the specification. Tests for all inspection requirements shall be conducted in accordance with table I. Table I - Index is shown below. The adhesion strength to substrate shall be performed for class 1 (300 C), class II (500 C), class III (1000 C), and class IV (1500 C) in accordance with ASTM tape test D3359 at room temperature. Visual inspection or examination shall be conducted in accordance with the contract or purchase order. Inspection for determination of compliance with 4.4.2 shall be made as specified by the procuring activity. Elevated temperature test schedule shall be conducted in accordance with ASTM D2485, section 7.3 / Test Method B - 7.3.1, 7.3.2, and 7.3.3 - visual inspection paragraphs.

Table I. Index.

Item	Test Method	Test Paragraph	Requirement Paragraph
Electrical conductivity	ASTM B193, ASTM D4496, IPC-TM-650	4.7.2	3.9, 3.10
Percent change in electrical conductivity	---	4.7.3	3.9, 3.11
Adhesion strength to substrate	ASTM D3359, ASTM 4541 / ISO 4624	4.7.4	3.9, 3.12
Single lap shear	ASTM D 3163	4.7.5	3.9, 3.13

4.9.1 Quality control / quality assurance plan. The document shall be a supplier specified quality program evidence.

4.9.2 Adhesion. Prepare test specimens and after bake cured, the adhesion testing shall be performed as specified in ASTM D3359 and D4541 respectively and examine for compliance with 3.8.1.

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4.10 Drop tower - acceleration shock test. Testing at 25,000g or 50,000g or 75,000g or 100,000g (1 g is the acceleration due to gravity) acceleration levels using a minimum of four specimens, each containing four circuits, shall be conducted. These tests shall be performed on a drop tower. These acceleration levels are determined to account for almost any potential application. Acceleration magnitude shall be taken by consensus of two accelerometers mounted to the table in accordance with manufacturer's recommendations. Peak acceleration value is the primary parameter taken from acceleration data. Data acquisition shall be conducted at 1 MHz sampling rates for all parameters. Trace resistance shall be monitored in real time to understand trace degradation and identify failed circuits. High speed videos shall be captured as necessary.

4.11 Rejection and retest. Failure in any inspection will result in the rejection of the materials from large scale production. Rejected parts cannot be resubmitted for acceptance without written approval from the contracting activity. The application for resubmission will contain all details concerning previous rejections and measures taken to correct these deficiencies.

4.12 Safety Data Sheet (SDS). The SDS shall address all components of the feedstock/ink and be prepared in compliance with FED-STD-313. The SDS shall be included with each shipment of the material and forwarded to pertinent Government agencies as stated in FED-STD-313. Contracting officers will identify those activities requiring copies of a complete SDS and submission of data. Noncompliance to 3.16 shall be cause for rejection.

4.13 Manufacturer's certification. The manufacturer shall certify and maintain substantiating evidence (Objective Quality Evidence) that the ink meets the requirements of this specification, and conforms to the manufacturer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery or provided for under the provisions of the contract.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order. When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the person responsible for packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that is helpful but is not mandatory.)

6.1 Intended use. The conductive ink properties covered by this performance specification are application unique. They are intended for use, throughout the Department of Defense, on rigid

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and flexible substrates of additively manufactured components. All applications in accordance with this specification and all criteria deviation should be as designated in the purchasing documentation by the acquisition activity. This document is intended to increase awareness and usage of these materials for fabrication, their elevated temperature applications, quality plans, and customized end user requirements.

6.2. Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2.1 and 1.2.2).
c. If issues of documents are different (see 2.2.1 and 2.3).
d. Specify type of conductive ink required (see 1.2.1, 1.2.2, and 3.1).
e. If the first article inspection and submission are different (see 3.6 and 4.2).
f. If specific conductive requirements are required (see 3.7).
g. If a special quality level in the quality plan is required (see 3.8).
h. Additional surface trace requirements (see 3.14).
i. Selection of applicable level of markings and packaging requirements (see 3.15 and 5.1).
j. If required more than safety data sheet, any additional relevant data sheets, health and safety (3.16 and 4.12).
k. When certification is acceptable (see 4.3 and 6.10).
l. Special tests as specified by the procuring authority (see 4.7).
m. Other requirements or special conditions other than those listed (see 4.8).
n. Packaging requirements if other than specified (see 5.1).

6.3 Objective Quality Evidence. A documented statement of fact, or supporting record, either quantitative or qualitative, pertaining to the quality of an item or process, based on observations, measurements, or tests that can be verified as required within the document. Evidence will be expressed in terms of specific quality requirements or characteristics in a certified lab report. These characteristics are identified in drawings, specifications, and other documents, which describe the item, process, or procedure.

6.4 Basis of purchase. Container units by volume are furnished as a single component in bottles or tubes (cartridges). All containers should be sealed to prevent moisture intrusion. A product in meeting the applicable requirements of this specification should result (see 6.2).

6.5 ASTM F2792 and IPC-T-51. The definitions standard - Standard Guide for Printed Hybrid Electronics for terms that are used in electronics industry. Reference should be made to Terms and Definitions for the Design and Manufacture of Printed Electronics - IPC T-51 for quality control/assurance and it is available for terminology, application and inspection language for conductive inks.

6.6 SAE Government Electronics and Information Technology Association, GEIA-859. This data management standard should be available for any potential application. And SAE Aerospace

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Material Specification, AMS-7002A, AMS-7003, and AMS-7032 process standards should be used if appliable.

6.7 Cognizant Engineering Authority. The CEA assumes all responsibility for assuring that all required testing and subsequent evaluation of any inks are completed and pass all requirements specified herein, and also meet any additional performance criteria specific to their intended application.

6.8 Engineering Support Activity. The ESA provides technical data, quantities, packaging requirements, possible suppliers, special handling, storage requirements, and shelf-life information to procure the ink and resolve individual engineering support problems.

6.9 Process Approval. When specified in the contract (see 4.3 and 6.2), for types I and II and classes I - IV, in lieu of submitting written procedures identifying processes, suppliers certified as specified herein are acceptable. Be prepared to offer evidence of the certification if requested.

a. National Aerospace and Defense Contractors Accreditation program (NADCAP), Performance Review Institute (PRI) industry managed accreditation programs. See https://www.eauditnet.com.

b. NACE Business Services, LLC, NACE International Institute for Contractor Accreditation Program (NIICAP) accredited contractors. See https://www.niicap.net/.

c. ISO 9001/2015 or ISO/IEC 17025 Quality Management, Conformity Assessment Accreditation from an external certification body. See: https://www.iso.org/home.html.

In reviewing and validating certification by the Engineering Support Activity (ESA), suppliers certified should submit their accreditation from the external certification body or registrar.

6.10 Institute for Printed Circuits (IPC) Standard Certification Program. The certification levels include Specialist, Expert, and Trainer. The Program is given at Authorized Certification Centers by the Technical Training Committee.

6.11 Lansmont Drop Tower (LDT). The LDT is used to bring about shock loads to printed ink and can achieve acceleration level of 30,000 g.

6.12 MIL-HDBK-454C. Helpful information may be obtained from Area: SESS, “General Guidelines for Electronic Equipment”, 21 September 2021, Guideline 17, Printed Wiring, page 17-1.

6.13 Additive Fabrication. Method, material, equipment, atmosphere, and manufacturing processes should be specified by the procuring activity.

6.14 ASTM F3456 standard. This standard is for users of polymer and metal powders. The document is an addition as a starting point for reuse schema for ink materials, and to develop appropriate practices and regulatory limitations prior to use.

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6.15 Toxicity clearance request. Department of the Army Regulation (AR) 40-5, Preventive Medicine, (AR) 70-1, Acquisition Policy, and Department of the Army Pamphlet 70-3, Acquisition Procedures, require toxicity clearance. Army toxicity questions and/or a toxicity clearance request should be sent to: US Army Public Health Command (USAPHC). The USAPHC point of contact is currently the Defense Center for Public Health (DCPH-A): DHA Public Health, ATTN: DCPH-ATS-TOX, 8300 Ricketts Point Road, APG-EA, MD 21010-5403 or emailed to:

dha.apg.Pub-Health-A.mbx.tox-info@health.mil.

6.16 Performance specifications. MIL-PRF-32667 feedstock material publication (document date: 28 October 2020), Ultra High Strength Steel Powder for Additive Manufacturing Fabrication of Parts, is available for sourcing the steel alloy powder and quality control inspection to establish a successful build. MIL-PRF-32753, Finished Material Properties for Additive Manufactured Fabrication of Low Alloy Steel Components (document date: 14 September 2023) is available for the minimum mechanical property requirements.

6.17 Subject term (key word) listing.

Adhesion	Fabrication	Service
Certification	Heat treatment	Sintering
Component	Minimum acceptance	Substrates
Dielectric	Parts	Surface
Electrical	Quality	Trace

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - AS

Preparing Activity:
Army - MR

(Project 9630-2025-001)

Review activities:
Army - AV, CR
Navy - OS, SH
DLA - CQ, DH

Civil agency:
GSA - FAS

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